

RESEARCH LETTER

10.1002/2017GL076005

Key Points:

- Groundwater sustainability metrics require more than a trend in groundwater storage changes
- A sustainability framework using a GRACE-groundwater drought index provides a robust time series to evaluate groundwater sustainability

Supporting Information:

- Supporting Information S1

Correspondence to:

B. F. Thomas,
bftthomas@pitt.edu

Citation:

Thomas, B. F., Caineta, J., & Nanteza, J. (2017). Global assessment of groundwater sustainability based on storage anomalies. *Geophysical Research Letters*, 44, 11,445–11,455. <https://doi.org/10.1002/2017GL076005>

Received 12 OCT 2017

Accepted 8 NOV 2017

Accepted article online 13 NOV 2017

Published online 27 NOV 2017

©2017. The Authors.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Global Assessment of Groundwater Sustainability Based On Storage Anomalies

Brian F. Thomas¹ , Júlio Caineta^{1,2}, and Jamiat Nanteza³ 
¹Department of Geology and Environmental Science, University of Pittsburgh, Pittsburgh, PA, USA, ²Computational Modeling and Simulation Program, Department of Geology and Environmental Science, University of Pittsburgh, Pittsburgh, PA, USA, ³Department of Geography, Geo-Informatics and Climate Sciences, Makerere University, Kampala, Uganda

Abstract The world's largest aquifers are a fundamental source of freshwater used for agricultural irrigation and to meet human water needs. Therefore, their stored volume of groundwater is linked with water security, which becomes more relevant during periods of drought. This work focuses on understanding large-scale groundwater changes, where we introduce an approach to evaluate groundwater sustainability at a global scale. We employ a groundwater drought index to assess performance metrics (reliability, resilience, vulnerability, and a combined sustainability index) for the largest and most productive global aquifers. Spatiotemporal changes in total water storage are derived from remote sensing observations of gravity anomalies, from which the groundwater drought index is inferred. The results reveal a complex relationship between the indicators, while considering monthly variability in groundwater storage. Combining the drought and sustainability indexes, as presented in this work, constitutes a measure for quantifying groundwater sustainability. This framework integrates changes in groundwater resources due to human influences and climate changes, thus opening a path to assess progress toward sustainable use and water security.

1. Introduction

Alarming observations in the recent decline of groundwater storage (Döll et al., 2014; Richey, Thomas, Lo, Reager, et al., 2015) coincide with an estimated 70% of freshwater withdrawals being used for irrigated agriculture (Rosegrant et al., 2009). Reliance on groundwater to fulfill agriculture water demands can be regarded as a consequential need for food security (Carruthers et al., 1997), thus linking water scarcity as a future constraint to food production (UN, 2007). Groundwater depletion strategies are recognized as a challenge to mitigate water scarcity (Rosegrant & Cline, 2003), yet uncertainty regarding the influence of climate change and groundwater use remain. During drought, agricultural water demands are often fulfilled by resilient groundwater storage (Castle et al., 2014; Famiglietti, 2014; Famiglietti et al., 2011), potentially altering aquifer resilience (Rodell et al., 2009; University of California Center for Hydrologic Modeling, 2014) and hydraulic properties including aquifer compaction (Smith et al., 2017). It remains unclear, however, how large-scale groundwater sustainability may be assessed to capture complex groundwater relationships.

Remote sensing has been established as a powerful tool to observe water storage dynamics at large scales (Chen et al., 2016; Long et al., 2017; Richey, Thomas, Lo, Reager, et al., 2015; Rodell & Famiglietti, 1999; Schmidt et al., 2006; Wahr et al., 2006) with the launch of the Gravity Recovery and Climate Experiment (GRACE) satellites. Previous applications of GRACE have isolated groundwater storage changes (Castle et al., 2014; Famiglietti et al., 2011; Nanteza et al., 2016; Rodell et al., 2009; Scanlon, Faunt, et al., 2012; Scanlon, Longuevergne, et al., 2012) to improve our understanding of regional groundwater storage changes. To date, global groundwater assessments have evaluated trends of groundwater storage (Richey, Thomas, Lo, Reager, et al., 2015) or have combined hydrologic model output with observations to identify the influence of groundwater abstractions to groundwater trends (Döll et al., 2014). Here we apply the normalized GRACE-groundwater drought index (GGDI) to quantify a sustainability index as a function of performance indicators for the largest global aquifers (Margat and Van der Gun, 2013) to recognize the consequence of temporal aquifer storage changes. Despite the temporal (Gleeson, Alley, et al., 2012) and spatial (Thomas & Famiglietti, 2015) coverage requirements for sustainability assessments that analyses derived through GRACE cannot satisfy (Alley & Konikow, 2015), the approach introduced here seeks to further our understanding of large-scale aquifer behaviors as a response of human- and climate-driven influences.

Various sustainability metrics have been applied to both groundwater and surface water systems to identify critical water security regions (Gain et al., 2016; Vrba & Zaporozec, 1994; Wada et al., 2010; White, 2013). The application of performance indicators such as reliability, resilience, and vulnerability is common in hydrology (Ajami et al., 2008; Asefa et al., 2014; Kjeldsen & Rosbjerg, 2004; Sandoval-Solis et al., 2010) to characterize stochastic performance of water resources systems. Performance indicators quantify system measures based on subjective a priori determined criteria (i.e., can a reservoir storage meet water demand?). Despite these limitations, the application of performance indicators remains attractive since they may incorporate various criteria (Loucks, 1997) as compared to evaluating a single criterion.

Although performance indicators have long been used to evaluate surface water systems, their application to groundwater has been limited (Hirata et al., 2007; Mays, 2013; Peters et al., 2005; Van Camp et al., 2010; Vrba et al., 2007). Applying performance indicators to large-scale aquifers (Margat & Van der Gun, 2013) allows for holistic assessments of groundwater sustainability. Our approach highlights complex relationships of groundwater sustainability where competing factors may influence one indicator (e.g., reliability) and not influence a second indicator (e.g., vulnerability). The results indicate a complex relationship between performance indicators, highlighting that this methodology yield different conclusions than previous storage-based analyses.

2. Methods

The assessment of groundwater sustainability is conducted for the largest aquifers in the Worldwide Hydrogeological Mapping and Assessment Program (WHYMAP, (Margat & Van der Gun, 2013)), known widely as the "Large Aquifer Systems of the World." These aquifers comprise the world's most productive groundwater systems that contain significant global resources (Margat, 2007; Richey, Thomas, Lo, Famiglietti, et al., 2015) to fulfill irrigation and drinking water demands. These aquifers are of interest as each fulfills spatial resolution requirements for GRACE observations (i.e., greater than 150,000 km² (Rodell & Famiglietti, 1999)).

2.1. GRACE Observations

Remote sensing observations from the GRACE satellites are used to quantify changes in groundwater storage in the study aquifers. GRACE observes monthly gravity anomalies which can be converted to monthly changes in terrestrial water storage (Rodell & Famiglietti, 1999). Monthly GRACE gravity coefficients (Tapley et al., 2004) from the JPL-RL05.01M Mascon solution (Watkins et al., 2015; Wiese, 2015) over the period of April 2002 through July 2016 were used for this study. Average water storage changes for all aquifers were computed as anomalies of water storage relative to the baseline time period of January 2004 to December 2009. Groundwater storage changes are estimated using a mass balance approach whereby auxiliary data sets permit isolation of a groundwater storage signal from total water storage changes (Rodell & Famiglietti, 2002). We assume that the total water storage anomaly (TWSA) is composed of anomaly changes in soil moisture (SMA), snow water equivalent (SWEA), surface water/reservoir storage (SWA), and groundwater (GWA) (equation (1)).

$$\text{TWSA} = \text{SMA} + \text{SWEA} + \text{SWA} + \text{GWA} \quad (1)$$

Groundwater storage anomalies were estimated by rearranging equation (1), while errors for the GWA time series were estimated by propagating TWSA, SMA, SWA, and SWEA error (Rodell & Famiglietti, 2002). We used monthly output from four models within the NASA Global Land Data Assimilation System (GLDAS, (Rodell et al., 2004)), including Noah (Chen et al., 1996; Koren et al., 1999), VIC (Liang et al., 1996), CLM2.0 (Dai et al., 2003), and Mosaic (Koster & Suarez, 1992), to remove hydrologic components from TWSA using equation (1) (Rodell & Famiglietti, 2002) (see supporting information). Great uncertainty in isolating groundwater storage from TWSA has been documented (Döll et al., 2012; Güntner et al., 2007; Swenson & Lawrence, 2015). In our evaluation, combinations of mass balance components from GLDAS models were tested to account for SMA (at depths of both 0–1 m and 0–2 m), SWA, and SWEA in equation (1). Given the models used and variables of interest, a total of 128 realizations of GWA were calculated (see supporting information), to account for uncertainty in isolating GWA.

2.2. GRACE Groundwater Drought Index

Thomas et al. (2017) introduced the GRACE groundwater drought index (GGDI), a normalized groundwater storage-based indicator to evaluate groundwater drought for large-scale aquifer systems. The novelty of

GGDI is represented by incorporation of groundwater storage, both as a surplus or deficit (Thomas et al., 2014), to permit an evaluation of groundwater drought. GGDI was derived from GWA for all realizations.

In brief, after isolating GWA, we estimate GGDI by first calculating a monthly climatology (C_i , the climatology for month i , Thomas et al., 2014), as in equation (2).

$$C_i = \frac{\sum_1^{n_i} \text{GWA}_i}{n_i} \text{ where } i = 1, \dots, 12 \quad (2)$$

In equation (2), GWA_i represents the GRACE-derived groundwater storage anomaly for month i and n represents the number of GWA estimates for month i . The monthly climatology removes the influence of seasonality in groundwater storage changes (Jasechko et al., 2014). The computed monthly climatology is removed from GWA to obtain a groundwater storage deviation (GSD), which represents the net deviation in the volume of groundwater storage based on seasonality. Finally, we normalize the GSD by removing the mean ($\bar{x}_{\text{GSD}}$), which is essentially zero as GWA represents an anomaly, and dividing by the standard deviation (s_{GSD}) as in equation (3).

$$\text{GGDI} = \frac{\text{GSD}_t - \bar{x}_{\text{GSD}}}{s_{\text{GSD}}} \quad (3)$$

Thus, GGDI represents the normalized net deviation in groundwater storage in all study aquifers.

2.3. Performance Indicators

Performance indicators including reliability, resilience, and vulnerability are applied to each of the WHYMAP aquifers. These performance indicators are typically applied using weighted, multiobjective approaches to characterize stochastic performance metrics of water resources systems. As applied here, we assume only a single planning period represented by an average of GGDI from all realizations. Further, performance metrics are classically applied given a relation between demand and storage (Hashimoto et al., 1982; Loucks et al., 1981). For this analysis, we apply a weighted statistical indices scheme (Loucks, 1997; Mays, 2013) to GGDI. Although the selection of satisfactory and unsatisfactory conditions is subjective, we assign the threshold as a function of the normalized GGDI: a positive GGDI is satisfactory and a negative GGDI is unsatisfactory. We adhere to a 3 month condition as proposed by Thomas et al. (2017) in the application of GGDI by conducting the performance metric analysis on 3 month smoothed averages of GGDI. Lastly, it is important to note that although GGDI is normalized, the distribution of positive and negative values is not equal as GGDI is representative of storage change anomalies over the study period.

2.3.1. Reliability

Following Loucks (1997) and Mays (2013), we quantify reliability as

$$\text{REL} = \frac{\# \text{ satisfactory conditions}}{\text{total } \# \text{ conditions}} \quad (4)$$

where reliability is defined by how often a system fails (Hashimoto et al., 1982). When applied to large-scale aquifers, reliability represents the historical likelihood that the aquifer storage falls below a normal condition (Thomas et al., 2017).

2.3.2. Resilience

Following Loucks (1997) and Mays (2013), we quantify resilience as

$$\text{RES} = \frac{\# \text{ times a satisfactory condition follows an unsatisfactory condition}}{\text{total } \# \text{ unsatisfactory conditions}} \quad (5)$$

Resiliency indicates how quickly a system returns to a satisfactory condition after an unsatisfactory condition (Hashimoto et al., 1982). A recovery is scored if GGDI shifts from negative to positive. As applied to large-scale aquifers, resilience represents the likelihood of a return from unsatisfactory to satisfactory conditions.

2.3.3. Vulnerability

We define vulnerability as a probabilistic measure that accounts for the extent and magnitude of failure (Hashimoto et al., 1982), where failure is synonymous with unsatisfactory. Given this definition, we must

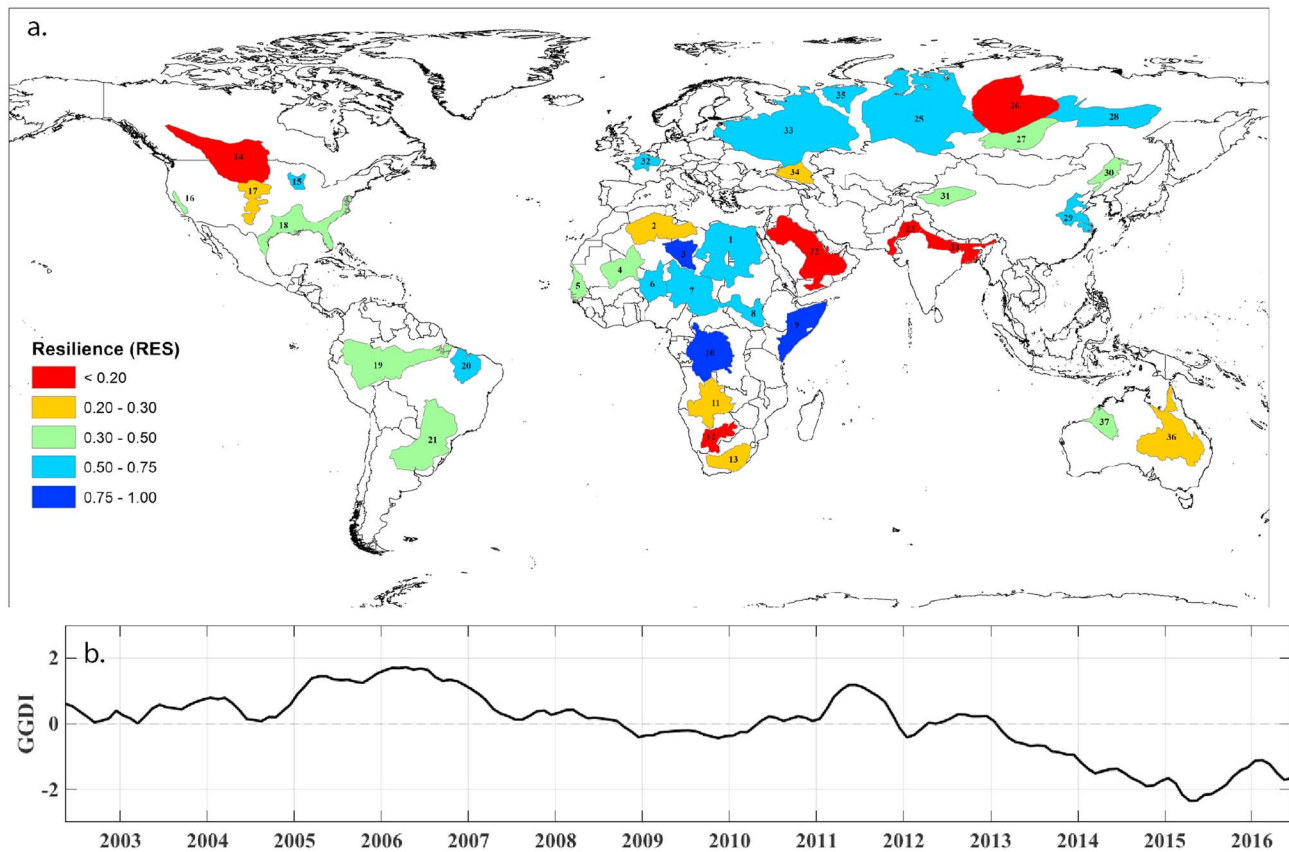


Figure 1. (a) Resilience of the largest aquifers (cf. WHYMAP) and (b) the average GGDI time series for the Central Valley Aquifer (16). The aquifer response after the 2007 drought resulted in a resilience score of 0.32.

account for both the magnitude of the event ($s_j = \text{GGDI}_j$) and the probability of the severity of the magnitude (e_j), during the study period, F , where

$$\text{VUL} = \sum_{j \in F} s_j e_j \quad (6)$$

This approach magnifies the influence of extreme drought conditions observed during the study period, conditions documented to greatly influence water security (Famiglietti & Rodell, 2013).

3. Results

3.1. GGDI

A comparison of GGDI estimates (Figures S1b–S37b) illustrates less variability among the 128 realizations, computed from various combinations of GLDAS model output, as compared to estimates of GWA (Figures S1a–S37a). For instance, the lowest normalized root-mean-square error (nRMSE) between the mean time series for GGDI was 0.38 (Atlantic and Gulf Coastal Plains Aquifer (18)), while the minimum nRMSE for GWA was -1.22 (Atlantic and Gulf Coastal Plains Aquifer (18)).

The influence of normalizing GWA to derive GGDI negates the influence of inconsistent sources of auxiliary data to isolate a representative groundwater signal (Reager et al., 2015). Additionally, as shown in previous GRACE studies, uncertainties in the isolated GWA signal are influenced by multiple factors, including land surface model deficiencies (Rodell et al., 2009; Rodell & Famiglietti, 2001), spatial resolution, and filtering errors (Longuevergne et al., 2013; Nanteza et al., 2016). Furthermore, isolation of drought indicators has been shown to be influenced by deep soil moisture and groundwater (Houborg et al., 2012; Rodell & Famiglietti, 2001; Swenson & Lawrence, 2015), since deep soil moisture introduces persistence into model simulations (Sheffield & Wood, 2007).

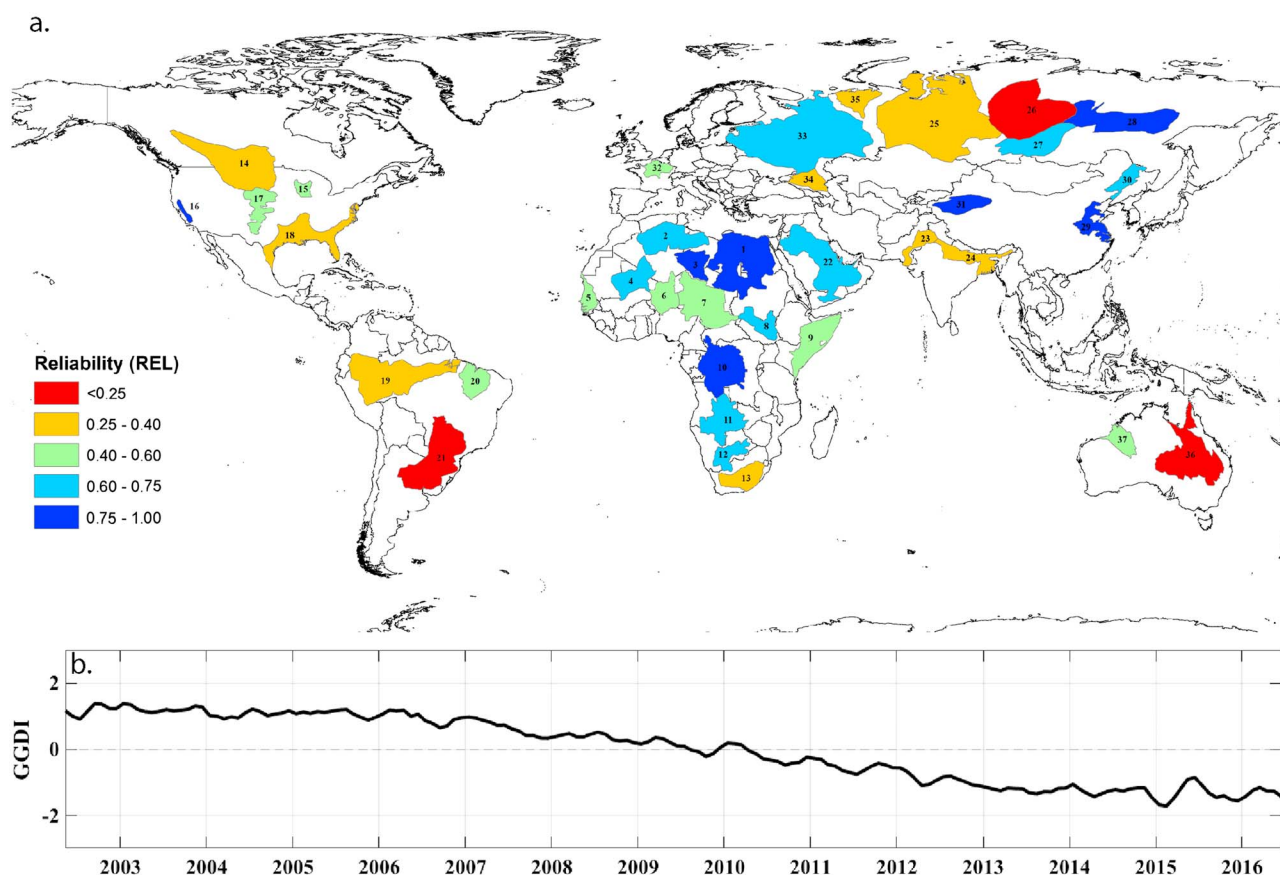


Figure 2. (a) Reliability of the largest aquifers (cf. WHYMAP) and (b) the average GGDI time series for the Arabian Aquifer (22). In the Arabian aquifer, a lack of variability in GGDI resulted in a high reliability score.

3.2. Performance Indicators

The magnitude of the performance indicators is a function of the selection of satisfactory and unsatisfactory conditions (see supporting information). We have normalized the results of performance indicators and of the sustainability index from 0 to 1, based on the maximum and minimum values for each indicator to permit aquifer comparisons. Numerical aquifer keys, as summarized in Table S1 and in figures, are listed after each described aquifer.

3.2.1. Resilience

Results of our quantification of resilience are illustrated in Figure 1a. It is apparent that aquifers in arid regions, that is, the Northwestern Sahara Aquifer System (2) and the Arabian Aquifer System (22), exhibit low resilience. These two aquifers are considered nonrenewable (Margat & Van der Gun, 2013) and have been characterized as overstressed (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015). Aquifers that exhibit higher resilience are found in high precipitation regions and northern latitudes (West Siberian Basin (25), Russian Platform Basins (33), and Congo Basin (10)). These aquifers may be considered resilient as a result of many factors, including relations between recharge and precipitation, whereby exposed bedrock aquifers may be recharged rapidly after dry periods. Many of the highly resilient aquifers were characterized as variable stressed (e.g., the Congo Basin (10), the Russian Platform Basins (33), and the Paris Basin (32)) or overstressed (e.g., the Nubian Aquifer System (1), the Murzuk-Djado Basin (3), and the Ogaden-Juba Basin (9)) by Richey, Thomas, Lo, Reager, et al. (2015). The Central Valley aquifer (16) exhibited medium resilience ($RES = 0.32$) due to inefficient drought recovery in the 2011 drought period, as compared to the 2007 drought period (Figure 1b), as identified by Thomas et al. (2017). A lower resilience score for the Central Valley is aggravated due to extensive groundwater withdrawals and rapid groundwater depletion known to occur during periods of drought (Famiglietti et al., 2011; Faunt et al., 2016).

3.2.2. Reliability

Aquifers exhibiting high reliability (Figure 2a) often feature large storage (Richey, Thomas, Lo, Famiglietti, et al., 2015). For example, the Arabian Aquifer System (22) is the world's largest and correspondingly is one of the most reliable aquifers. The Nubian Aquifer System (1) also exhibited high reliability as a result of its large storage, despite little modern-day recharge. The Central Valley Aquifer (16) exhibited high reliability, and though this result seems counterintuitive given historic overdraft of the aquifer (Famiglietti et al., 2011; Faunt, 2009), it is important to recognize that GGDI is a function of GRACE storage changes and is impacted by drought (Thomas et al., 2017).

In a comparison to groundwater stress characterization, reliable aquifers tend to be characterized as over-stressed aquifers (e.g., the Arabian Aquifer System (22), the Nubian Aquifer System (1), the Northwestern Sahara Aquifer System (2), and the Murzuk-Djado Basin (3) (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015)), while less reliable aquifers tend to be characterized as variably stressed or unstressed. Reliable aquifers exhibit such behaviors as a result of gradual decreases in groundwater storage which will effectively increase GGDI early in the time series (Figure 2b). Variably stressed aquifers may be impacted by declines in water quality due to increased induced recharge (Theis, 1940; Richey, Thomas, Lo, Reager, et al., 2015) thus underscoring concern regarding future water quality.

3.2.3. Vulnerability

As depicted in Figure 3a, the least vulnerable aquifers are found throughout equatorial Africa and the northern latitudes of Europe and Asia. These aquifers are generally characterized as overstressed and variably stressed (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015), indicating that aquifer vulnerability is largely influenced by groundwater recharge. Nonrenewable aquifers including the Northwestern Sahara Aquifer System (2) and Arabian Aquifer system (22) also exhibited high vulnerability, further indicating a close relationship between aquifer vulnerability and groundwater recharge potential. The Northern Great Plains aquifer (14) and the Upper Kalahari-Culvelai-Upper Zambezi Basin (11), which share similar behavior in GGDI, exhibited the most vulnerability, a factor of prolonged droughts over the respective regions combined with rapid increases in subsurface storage (Hoerling et al., 2014; Thomas et al., 2014), which impacts the distribution of positive and negative values in GGDI (Figure 3b). Results depicted in Figure 3a deviate from reported groundwater stress regimes. The difference is attributed to our vulnerability calculation from a 3 month moving average, while renewable groundwater stress was based on GWA trends or simulated groundwater recharge (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015).

3.3. Sustainability Index

We calculate a sustainability index, which is a function of resilience, reliability, and vulnerability, as given by equation (7) (Loucks, 1997).

$$\text{sustainability index (SI)} = \text{REL} \times \text{RES} \times (1 - \text{VUL}) \quad (7)$$

The fact this index is multiplicative rather than additive contributes weight to the indicator having the lowest value. In the case that one of the indicators is zero, it is unlikely that any other indicator would be as relevant. This index could be further expanded to a relative index, by applying various weighting metrics, although such action increases subjective judgment (Loucks, 1997). As depicted in Figure 4, aquifers in the arid regions of the Saharan desert and Arabian Peninsula, in addition to aquifers in northern India, exhibit unsustainable groundwater characteristics. The Northwestern Sahara (2), Arabian (22), and Indus (23) aquifers were identified as overstressed (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015), further illustrating the importance of groundwater sustainability interventions for these aquifer systems. The Great Artesian (36) and Canning Basin (37) aquifers in Australia exhibited unsustainable groundwater characteristics. In those aquifers, SI was reduced due to low resilience (Figure 1a), a factor largely influenced by the prolonged drought that occurred over the region. The Central Valley aquifer (16) value of SI, 0.37, was impacted by low resilience. Similar to the Australian aquifers, the Central Valley was impacted by multiple drought periods (Thomas et al., 2017). The High Plains aquifer (17) received a SI score of 0.23. It is well known that the High Plains system has been pumped unsustainably for decades (McGuire, 2009; Scanlon, Faunt, et al., 2012). However, during the GRACE record, the High Plains aquifer has benefited from active management strategies. This is especially true for the northern High Plains (Richey et al., 2016; Sahoo et al., 2016), where groundwater storage is observed to increase. The inability to separate the Northern High Plains from the Southern High Plains illustrates an important limitation of this study, where results are valid only at large scale (Alley

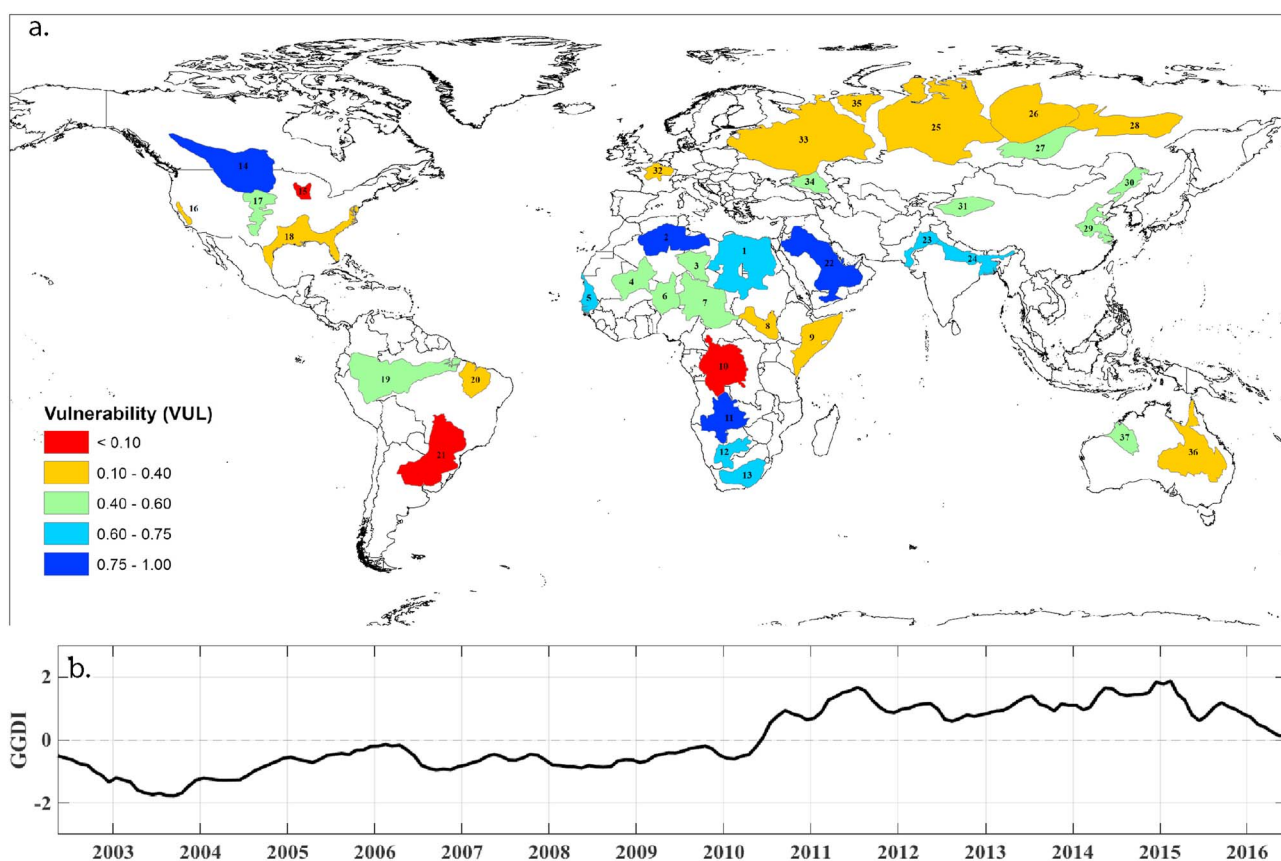


Figure 3. (a) Vulnerability of the largest aquifers (cf. WHYMAP) and (b) the average GGD time series for the Northern Great Plains Aquifer (14). The Northern Great Plains aquifer received a high vulnerability score due to GGD changing from negative up to 2010, and positive thereafter.

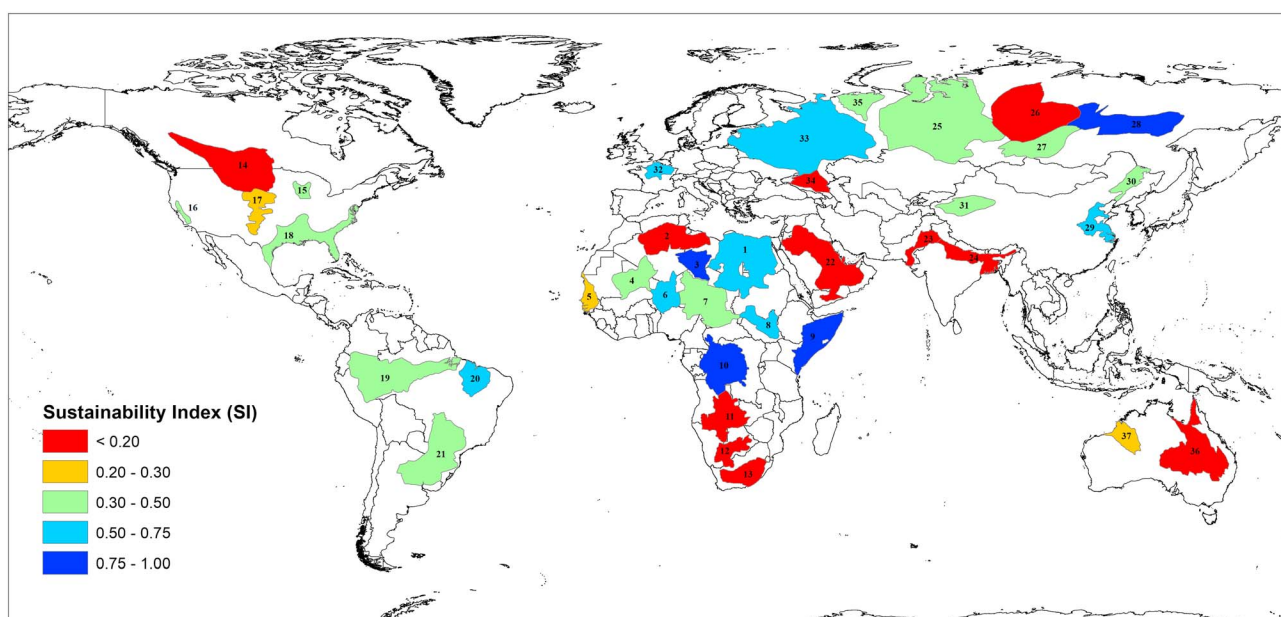


Figure 4. Sustainability index of the largest aquifers (cf. WHYMAP).

& Konikow, 2015). Although our results ignore spatial variability in groundwater recharge and abstractions that are important to characterize sustainability (Thomas & Famiglietti, 2015), Figure 4 highlights large-scale sustainability observed by GRACE.

4. Discussion

Previous global estimates of groundwater stress are hindered by inconsistent assumptions and differing estimates of use and availability (Döll, 2009; Gleeson, Wada, et al., 2012; L'vovich, 1979; Richey, Thomas, Lo, Reager, et al., 2015; Vörösmarty et al., 2000; Wada et al., 2010), thus raising important concerns regarding the evaluation of groundwater sustainability. Although groundwater serves as the primary source for agriculture water (Rosegrant et al., 2009), the importance of groundwater is rapidly increasing, given groundwater storage serves as a resilient supply as surface water becomes less reliable and unpredictable (Kundzewicz & Döll, 2009). Our results illustrate that capturing aquifer response may be extended beyond a single metric to holistically evaluate the relation of aquifer changes through various performance indicators. Results presented here integrate changes in groundwater resources as a function of human pressures and groundwater storage changes linked to climate and natural variability, as captured by GRACE, thus fulfilling challenges to assess development goals for achieving sustainability advanced by the United Nations (Connor, 2015).

Performance indicators including resilience, reliability, and vulnerability were applied to a GRACE-based groundwater drought index, GGDI (Thomas et al., 2017), for the "Large Aquifer Systems of the World" (Margat and Van der Gun, 2013). Our results illustrate that, globally, aquifer dynamics characterized by GRACE trends or groundwater stress (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015) fail to capture diverse factors which contribute to groundwater storage changes (Figure 4). For example, the Great Artesian Basin in Australia was noted to be unstressed in previous studies (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015). In our analysis, the aquifer exhibited low resilience and low reliability. We designated this aquifer as unsustainable as a result of the combination of aquifer metrics. The difference between our result and the previously identified unstressed conditions is that our sustainability analysis accounts for monthly variability. A comparison of performance indicators, SI, stress condition, and groundwater storage trends (Gleeson, Wada, et al., 2012; Richey, Thomas, Lo, Reager, et al., 2015) is summarized in Table S1. Table S1 indicates the variability in aquifer changes that effect the value of performance indicators, and hence SI, that are not accounted for in anomaly trends or groundwater stress. Our results capture integrated assessments of aquifer dynamics, thus providing a framework for future evaluations of aquifer sustainability.

5. Conclusions

Previous groundwater sustainability studies highlight the need to quantify sustainability metrics (Pandey et al., 2011; Gleeson, Alley, et al., 2012). To date, global groundwater assessments have applied stress indices (Richey, Thomas, Lo, Reager, et al., 2015) or have evaluated the influence of groundwater withdrawals in global groundwater storage (Döll et al., 2014; Gleeson, Wada, et al., 2012). This study has shown that temporal changes in groundwater storage influence the complex interactions in global aquifer systems, thus resulting in an integrated assessment of groundwater sustainability. Our use of GRACE to identify changes in groundwater storage captures the influence of groundwater use, the aquifer response to withdrawals via capture and natural variability. The combination of GGDI and performance metrics is proposed as a means to quantify groundwater sustainability and could be used as a tool to evaluate changes in observed groundwater storage changes over time to monitor sustainability in these systems.

The results of the study highlight the complex interactions resulting in an integrated assessment of groundwater sustainability. Previous studies have documented alarming changes in groundwater storage (Famiglietti et al., 2011; Rodell et al., 2009). Unlike those previous studies whose results are based on a single trend of groundwater storage changes, the analysis presented uses a 3-month average time series to characterize groundwater sustainability. Although GGDI provides a robust time series for the analysis used, it is important to recognize the limitations of the GRACE observations used to derive GGDI. For example, GRACE observes changes in one-dimensional groundwater storage, while important aquifer processes occur in three dimensions (Alley & Konikow, 2015). Further, the influence of soil moisture in isolation of a representative GRACE-derived groundwater storage signal is ambiguous (Swenson & Lawrence, 2015), despite further

processing to obtain GGDl. It is also relevant to note that the concept of vulnerability used in this work differs from the common denotation in the literature that accounts for the degree of potential susceptibility of groundwater to a pollution source. A possible link with water quality shall be developed in further studies.

Mays (2013) promoted multiobjective approaches to evaluate groundwater sustainability that include economic factors in addition to representations of aquifer changes. Although the analysis presented here does not account for variables other than groundwater storage, the results expand our understanding of large-scale groundwater processes by applying remotely sensed observations to document groundwater sustainability. Future research should seek to apply economic factors to the global analysis as presented here. More advanced approaches using multiobjective schemes may provide a broader understanding of performance indicators, although such approaches are likely restricted by the availability of groundwater data globally (Famiglietti et al., 2015).

Acknowledgments

This study was also made possible using freely available data from the Global Land Data Assimilation System (<http://disc.sci.gsfc.nasa.gov/hydrology/data-holdings>). GRACE land data are available at <http://grace.jpl.nasa.gov>, supported by the NASA MEaSUREs Program. Additional data used in this study are documented in the supporting information. The authors express their appreciation to the Editor and reviewers whose comments contributed to substantial improvements to the original manuscript.

References

- Ajami, N. K., Hornberger, G. M., & Sunding, D. L. (2008). Sustainable water resource management under hydrological uncertainty. *Water Resources Research*, 44, W11406. <https://doi.org/10.1029/2007WR006736>
- Alley, W. M., & Konikow, L. F. (2015). Bringing GRACE down to Earth. *Groundwater*, 53(6), 826–829. <https://doi.org/10.1111/gwat.12379>
- Asefa, T., Clayton, J., Adams, A., & Anderson, D. (2014). Performance evaluation of a water resources system under varying climatic conditions: Reliability, resilience, vulnerability and beyond. *Journal of Hydrology*, 508, 53–65. <https://doi.org/10.1016/j.jhydrol.2013.10.043>
- Carruthers, I., Rosegrant, M. W., & Seckler, D. (1997). Irrigation and food security in the 21st century. *Irrigation and Drainage Systems*, 11(2), 83–101. <https://doi.org/10.1023/A:1005751232728>
- Castle, S. L., Thomas, B. F., Reager, J. T., Rodell, M., Swenson, S. C., & Famiglietti, J. S. (2014). Groundwater depletion during drought threatens future water security of the Colorado River Basin. *Geophysical Research Letters*, 41, 5904–5911. <https://doi.org/10.1002/2014GL061055>
- Chen, J., Famiglietti, J. S., Scanlon, B. R., & Rodell, M. (2016). Groundwater storage changes: Present status from GRACE observations. *Surveys in Geophysics*, 37(2), 397–417. <https://doi.org/10.1007/s10712-015-9332-4>
- Chen, F., Mitchell, K., Schaake, J., Xue, Y., Pan, H. L., Koren, V., ... Betts, A. (1996). Modeling of land surface evaporation by four schemes and comparison with FIFE observations. *Journal of Geophysical Research*, 101(D3), 7251–7268. <https://doi.org/10.1029/95JD02165>
- Connor, R. (2015). *The United Nations world water development report 2015: Water for a sustainable world* (Vol. 1). France: UNESCO Publishing.
- Dai, Y., Zeng, X., Dickinson, R. E., Baker, I., Bonan, G. B., Bosilovich, M. G., ... Oleson, K. W. (2003). The common land model. *Bulletin of the American Meteorological Society*, 84(8), 1013–1023. <https://doi.org/10.1175/BAMS-84-8-1013>
- Döll, P. (2009). Vulnerability to the impact of climate change on renewable groundwater resources: A global-scale assessment. *Environmental Research Letters*, 4(3), 035006. <https://doi.org/10.1088/1748-9326/4/3/035006>
- Döll, P., Hoffmann-Dobrev, H., Portmann, F. T., Siebert, S., Eicker, A., Rodell, M., ... Scanlon, B. R. (2012). Impact of water withdrawals from groundwater and surface water on continental water storage variations. *Journal of Geodynamics*, 59, 143–156.
- Döll, P., Mueller Schmied, H., Schuh, C., Portmann, F. T., & Eicker, A. (2014). Global-scale assessment of groundwater depletion and related groundwater abstractions: Combining hydrological modeling with information from well observations and GRACE satellites. *Water Resources Research*, 50, 5698–5720. <https://doi.org/10.1002/2014WR015595>
- Famiglietti, J. S. (2014). The global groundwater crisis. *Nature Climate Change*, 4(11), 945–948.
- Famiglietti, J. S., Lo, M., Ho, S. L., Bethune, J., Anderson, K. J., Syed, T. H., ... Rodell, M. (2011). Satellites measure recent rates of groundwater depletion in California's Central Valley. *Geophysical Research Letters*, 38, L03403. <https://doi.org/10.1029/2010GL046442>
- Famiglietti, J. S., & Rodell, M. (2013). Water in the balance. *Science*, 340(6138), 1300–1301.
- Famiglietti, J. S., Cazenave, A., Eicker, A., Reager, J. T., Rodell, M., & Velicogna, I. (2015). Satellites provide the big picture. *Science*, 349(6249), 684–685.
- Faunt, C. C. (2009). *Groundwater availability of the central valley aquifer, California* (pp. 1–57). Reston, VA: US Geological Survey.
- Faunt, C. C., Sneed, M., Traum, J., & Brandt, J. T. (2016). Water availability and land subsidence in the Central Valley, California, USA. *Hydrogeology Journal*, 24(3), 675–684. <https://doi.org/10.1007/s10040-015-1339-x>
- Gain, A. K., Giupponi, C., & Wada, Y. (2016). Measuring global water security towards sustainable development goals. *Environmental Research Letters*, 11(12), 124015. <https://doi.org/10.1088/1748-9326/11/12/124015>
- Gleeson, T., Alley, W. M., Allen, D. M., Sophocleous, M. A., Zhou, Y., Taniguchi, M., & VanderSteen, J. (2012). Towards sustainable groundwater use: Setting long-term goals, backcasting, and managing adaptively. *Ground Water*, 50(1), 19–26. <https://doi.org/10.1111/j.1745-6584.2011.00825.x>
- Gleeson, T., Wada, Y., Bierkens, M. F., & van Beek, L. P. (2012). Water balance of global aquifers revealed by groundwater footprint. *Nature*, 488(7410), 197–200. <https://doi.org/10.1038/nature11295>
- Güntner, A., Stuck, J., Werth, S., Döll, P., Verzano, K., & Merz, B. (2007). A global analysis of temporal and spatial variations in continental water storage. *Water Resources Research*, 43, W05416. <https://doi.org/10.1029/2006WR005247>
- Hashimoto, T., Stedinger, J. R., & Loucks, D. P. (1982). Reliability, resiliency, and vulnerability criteria for water resource system performance evaluation. *Water Resources Research*, 18(1), 14–20. <https://doi.org/10.1029/WR018i001p00014>
- Hirata, R., Suhogusoff, A., & Fernandes, A. (2007). Groundwater resources in the State of São Paulo (Brazil): The application of indicators. *Anais da Academia Brasileira de Ciências*, 79(1), 141–152. <https://doi.org/10.1590/S0001-37652007000100016>
- Hoerling, M., Eischeid, J., Kumar, A., Leung, R., Mariotti, A., Mo, K., ... Seager, R. (2014). Causes and predictability of the 2012 Great Plains drought. *Bulletin of the American Meteorological Society*, 95(2), 269–282. <https://doi.org/10.1175/BAMS-D-13-00055.1>
- Houborg, R., Rodell, M., Li, B., Reichle, R., & Zaitchik, B. F. (2012). Drought indicators based on model-assimilated Gravity Recovery and Climate Experiment (GRACE) terrestrial water storage observations. *Water Resources Research*, 48, W07525. <https://doi.org/10.1029/2011WR011291>
- Jasechko, S., Birks, S. J., Gleeson, T., Wada, Y., Fawcett, P. J., Sharp, Z. D., ... Welker, J. M. (2014). The pronounced seasonality of global groundwater recharge. *Water Resources Research*, 50, 8845–8867. <https://doi.org/10.1002/2014WR015809>

- Kjeldsen, T. R., & Rosbjerg, D. (2004). Choice of reliability, resilience and vulnerability estimators for risk assessments of water resources systems/Choix d'estimateurs de fiabilité, de résilience et de vulnérabilité pour les analyses de risque de systèmes de ressources en eau. *Hydrological Sciences Journal*, 49(5). <https://doi.org/10.1623/hysj.49.5.755.55136>
- Koren, V., Schaake, J., Mitchell, K., Duan, Q. Y., Chen, F., & Baker, J. M. (1999). A parameterization of snowpack and frozen ground intended for NCEP weather and climate models. *Journal of Geophysical Research*, 104(D16), 19,569–19,585. <https://doi.org/10.1029/1999JD900232>
- Koster, R. D., & Suarez, M. J. (1992). Modeling the land surface boundary in climate models as a composite of independent vegetation stands. *Journal of Geophysical Research*, 97(D3), 2697–2715. <https://doi.org/10.1029/91JD01696>
- Kundzewicz, Z. W., & Döll, P. (2009). Will groundwater ease freshwater stress under climate change? *Hydrological Sciences Journal*, 54(4), 665–675. <https://doi.org/10.1623/hysj.54.4.665>
- Liang, X., Wood, E. F., & Lettenmaier, D. P. (1996). Surface soil moisture parameterization of the VIC-2L model: Evaluation and modification. *Global and Planetary Change*, 13(1–4), 195–206. [https://doi.org/10.1016/0921-8181\(95\)00046-1](https://doi.org/10.1016/0921-8181(95)00046-1)
- Long, D., Pan, Y., Zhou, J., Chen, Y., Hou, X., Hong, Y., ... Longuevergne, L. (2017). Global analysis of spatiotemporal variability in merged total water storage changes using multiple GRACE products and global hydrological models. *Remote Sensing of Environment*, 192, 198–216. <https://doi.org/10.1016/j.rse.2017.02.011>
- Longuevergne, L., Wilson, C., Scanlon, B. R., & Crétaux, J. F. (2013). GRACE water storage estimates for the Middle East and other regions with significant reservoir and lake storage. *Hydrology and Earth System Sciences*, 17(12), 4817–4830. <https://doi.org/10.5194/hess-17-4817-2013>
- Loucks, D. P. (1997). Quantifying trends in system sustainability. *Hydrological Sciences Journal*, 42(4), 513–530. <https://doi.org/10.1080/02626669709492051>
- Loucks, D. P., Stedinger, J. R., & Haith, D. A. (1981). *Water resource systems planning and analysis*. New York: Prentice-Hall.
- L'vovich, M. I. (1979). *World water resources and their future*. Washington, DC: American Geophysical Union. <https://doi.org/10.1029/SP013>
- Margat, J. (2007). Great Aquifer Systems of the World. In *Aquifer systems management: Darcy's Legacy in a world of impending water storage* (pp. 105–116). Boca Raton FL: CRC Press.
- Margat, J., & Van der Gun, J. (2013). *Groundwater around the world*. Leiden, Netherlands: CRC Press/Balkema.
- Mays, L. W. (2013). Groundwater resources sustainability: Past, present, and future. *Water Resources Management*, 27, 4409–4424. <https://doi.org/10.1007/s11269-013-0436-7>
- McGuire, V. L. (2009). Water-level changes in the High Plains aquifer, predevelopment to 2007, 2005–06, and 2006–07. Publications of the US Geological Survey, 17.
- Nanteza, J., de Linage, C. R., Thomas, B. F., & Famiglietti, J. S. (2016). Monitoring groundwater storage changes in complex basement aquifers: An evaluation of the GRACE satellites over East Africa. *Water Resources Research*, 52, 9542–9564. <https://doi.org/10.1002/2016WR018846>
- Pandey, V. P., Shrestha, S., Chapagain, S. K., & Kazama, F. (2011). A framework for measuring groundwater sustainability. *Environmental Science & Policy*, 14(4), 396–407. <https://doi.org/10.1016/j.envsci.2011.03.008>
- Peters, E., Van Lanen, H. A. J., Torfs, P. J. J. F., & Bier, G. (2005). Drought in groundwater—Drought distribution and performance indicators. *Journal of Hydrology*, 306(1–4), 302–317. <https://doi.org/10.1016/j.jhydrol.2004.09.014>
- Reager, J. T., Rao, P., Famiglietti, J. S., & Turmon, M. (2015). Improved uncertainty quantification in groundwater flux estimation using GRACE. In *AGU Fall Meeting Abstracts*.
- Richey, A. S., Thomas, B. F., Lo, M. H., Famiglietti, J. S., Swenson, S., & Rodell, M. (2015). Uncertainty in global groundwater storage estimates in a total groundwater stress framework. *Water Resources Research*, 51, 5198–5216. <https://doi.org/10.1002/2015WR017351>
- Richey, A. S., Thomas, B. F., Lo, M. H., Reager, J. T., Famiglietti, J. S., Voss, K., ... Rodell, M. (2015). Quantifying renewable groundwater stress with GRACE. *Water Resources Research*, 51, 5217–5238. <https://doi.org/10.1002/2015WR017349>
- Richey, A. S., Thomas, B. F., Lo, M. H., Reager, J. T., Famiglietti, J. S., Voss, K., ... Rodell, M. (2016). Reply to comment by Sahoo et al. on "Quantifying renewable groundwater stress with GRACE". *Water Resources Research*, 52, 4188–4192. <https://doi.org/10.1002/2015WR018329>
- Rodell, M., & Famiglietti, J. S. (1999). Detectability of variations in continental water storage from satellite observations of the time dependent gravity field. *Water Resources Research*, 35(9), 2705–2723. <https://doi.org/10.1029/1999WR900141>
- Rodell, M., & Famiglietti, J. S. (2001). An analysis of terrestrial water storage variations in Illinois with implications for the Gravity Recovery and Climate Experiment (GRACE). *Water Resources Research*, 37(5), 1327–1339. <https://doi.org/10.1029/2000WR900306>
- Rodell, M., & Famiglietti, J. S. (2002). The potential for satellite-based monitoring of groundwater storage changes using GRACE: The High Plains aquifer, Central US. *Journal of Hydrology*, 263(1–4), 245–256. [https://doi.org/10.1016/S0022-1694\(02\)00060-4](https://doi.org/10.1016/S0022-1694(02)00060-4)
- Rodell, M., Houser, P. R., Jambor, U. E. A., Gottschalk, J., Mitchell, K., Meng, C. J., ... Entin, J. K. (2004). The global land data assimilation system. *Bulletin of the American Meteorological Society*, 85(3), 381–394. <https://doi.org/10.1175/BAMS-85-3-381>
- Rodell, M., Velicogna, I., & Famiglietti, J. S. (2009). Satellite-based estimates of groundwater depletion in India. *Nature*, 460(7258), 999–1002. <https://doi.org/10.1038/nature08238>
- Rosegrant, M. W., & Cline, S. A. (2003). Global food security: Challenges and policies. *Science*, 302(5652), 1917–1919. <https://doi.org/10.1126/science.1092958>
- Rosegrant, M. W., Ringler, C., & Zhu, T. (2009). Water for agriculture: Maintaining food security under growing scarcity. *Annual Review of Environment and Resources*, 34(1), 205–222. <https://doi.org/10.1146/annurev.enviro.030308.090351>
- Sahoo, S., Russo, T., & Lall, U. (2016). Comment on "Quantifying renewable groundwater stress with GRACE" by Alexandra S. Richey et al. *Water Resources Research*, 52, 4184–4187. <https://doi.org/10.1002/2015WR018085>
- Sandoval-Solis, S., McKinney, D. C., & Loucks, D. P. (2010). Sustainability index for water resources planning and management. *Journal of Water Resources Planning and Management*, 137(5), 381–390.
- Scanlon, B. R., Faunt, C. C., Longuevergne, L., Reedy, R. C., Alley, W. M., McGuire, V. L., & McMahon, P. B. (2012). Groundwater depletion and sustainability of irrigation in the US High Plains and Central Valley. *Proceedings of the National Academy of Sciences of the United States of America*, 109(24), 9320–9325. <https://doi.org/10.1073/pnas.1200311109>
- Scanlon, B. R., Longuevergne, L., & Long, D. (2012). Ground referencing GRACE satellite estimates of groundwater storage changes in the California Central Valley, USA. *Water Resources Research*, 48, W04520. <https://doi.org/10.1029/2011WR011312>
- Schmidt, R., Schwintzer, P., Flechtner, F., Reigber, C., Güntner, A., Döll, P., ... Wünsch, J. (2006). GRACE observations of changes in continental water storage. *Global and Planetary Change*, 50(1–2), 112–126. <https://doi.org/10.1016/j.gloplacha.2004.11.018>
- Sheffield, J., & Wood, E. F. (2007). Characteristics of global and regional drought, 1950–2000: Analysis of soil moisture data from off-line simulation of the terrestrial hydrologic cycle. *Journal of Geophysical Research*, 112 D17115. <https://doi.org/10.1029/2006JD008288>
- Smith, R. G., Knight, R., Chen, J., Reeves, J. A., Zebker, H. A., Farr, T., & Liu, Z. (2017). Estimating the permanent loss of groundwater storage in the southern San Joaquin Valley, California. *Water Resources Research*, 53, 2133–2148. <https://doi.org/10.1002/2016WR019861>

- Swenson, S. C., & Lawrence, D. M. (2015). A GRACE-based assessment of interannual groundwater dynamics in the Community Land Model. *Water Resources Research*, 51, 8817–8833. <https://doi.org/10.1002/2015WR017582>
- Tapley, B. D., Bettadpur, S., Ries, J. C., Thompson, P. F., & Watkins, M. M. (2004). GRACE measurements of mass variability in the Earth system. *Science*, 305(5683), 503–505. <https://doi.org/10.1126/science.1099192>
- Theis, C. V. (1940). The source of water derived from wells. *Civil Engineering*, 10(5), 277–280.
- Thomas, B. F., & Famiglietti, J. S. (2015). Sustainable groundwater management in the arid Southwestern US: Coachella Valley, California. *Water Resources Management*, 29, 4411–4426. <https://doi.org/10.1007/s11269-015-1067-y>
- Thomas, B. F., Famiglietti, J. S., Landerer, F. W., Wiese, D. N., Molotch, N. P., & Argus, D. F. (2017). GRACE groundwater drought index: Evaluation of California Central Valley groundwater drought. *Remote Sensing of Environment*, 198, 384–392. <https://doi.org/10.1016/j.rse.2017.06.026>
- Thomas, A. C., Reager, J. T., Famiglietti, J. S., & Rodell, M. (2014). A GRACE-based water storage deficit approach for hydrological drought characterization. *Geophysical Research Letters*, 41, 1537–1545. <https://doi.org/10.1002/2014GL059323>
- UN (2007). Coping with water scarcity: Challenge of the twenty-first century. Prepared for World Water Day.
- University of California Center for Hydrologic Modeling (2014). UCCHM water advisory #1, water storage changes in California's Sacramento and San Joaquin River Basins from GRACE: Preliminary updated results for 2003–2013. Irvine, CA: University of California. Retrieved from <http://www.ucchm.org/publications>, last accessed 23 Dec. 2014.
- Van Camp, M., Radfar, M., & Walraevens, K. (2010). Assessment of groundwater storage depletion by overexploitation using simple indicators in an irrigated closed aquifer basin in Iran. *Agricultural Water Management*, 97(11), 1876–1886. <https://doi.org/10.1016/j.agwat.2010.02.006>
- Vörösmarty, C. J., Green, P., Salisbury, J., & Lammers, R. B. (2000). Global water resources: Vulnerability from climate change and population growth. *Science*, 289(5477), 284–288. <https://doi.org/10.1126/science.289.5477.284>
- Vrba, J., Girman, J., van der Gun, J., Haie, N., Hirata, R., Lopez-Gunn, E., ... Wallin, B. (2007). In A. Lipponen (Ed.), *Groundwater resources sustainability indicators* (p. 114). Paris: UNESCO.
- Vrba, J., & Zaporozec, A. (1994). Guidebook on mapping groundwater vulnerability. In *International contributions to hydrogeology* (Vol. 16). Heise, Hanover Germany.
- Wada, Y., van Beek, L. P., van Kempen, C. M., Reckman, J. W., Vasak, S., & Bierkens, M. F. (2010). Global depletion of groundwater resources. *Geophysical Research Letters*, 37, L20402. <https://doi.org/10.1029/2010GL044571>
- Wahr, J., Swenson, S., & Velicogna, I. (2006). Accuracy of GRACE mass estimates. *Geophysical Research Letters*, 33, L06401. <https://doi.org/10.1029/2005GL025305>
- Watkins, M. M., Wiese, D. N., Yuan, D. N., Boening, C., & Landerer, F. W. (2015). Improved methods for observing Earth's time variable mass distribution with GRACE using spherical cap mascons. *Journal of Geophysical Research: Solid Earth*, 120, 2648–2671. <https://doi.org/10.1002/2014JB011547>
- White, D. D. (2013). Framing water sustainability in an environmental decision support system. *Society & Natural Resources*, 26(11), 1365–1373. <https://doi.org/10.1080/08941920.2013.788401>
- Wiese, D. N. (2015). GRACE monthly global water mass grids NETCDF RELEASE 5.0. Ver. 5.0. PO.DAAC, CA, USA. Dataset accessed [2017–06–01] at <https://doi.org/10.5067/TEMSC-OCL05>